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Coursework Report

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MSc Risk Management and Financial Engineering

C2O - A cross-sectional, dollar-neutral overnight strategy

Machine Learning and Finance

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Abstract

This project studies whether U.S. equity close-to-open returns can be predicted and monetised using a cross-sectional machine learning strategy. I construct a point-in-time stock-date panel with adjusted close-to-open targets, annual top-1000 investable universes, capacity-aware eligibility filters, borrow-cost proxies, and approximately 250 features across fundamentals, earnings revisions, return-volatility-liquidity variables, technical indicators, and short-interest signals. The final model is a rolling random forest tail classifier with feature clustering, permutation filtering, PCA compression, validation-based hyperparameter selection, and out-of-sample prediction. The model achieves positive ranking ability, with an out-of-sample IC mean of 0.02318 and ICIR of 0.1561. However, after commission, auction slippage, and borrow costs, net performance becomes negative. The main conclusion is that the signal contains genuine predictive information, but the close-to-open implementation is structurally too cost-intensive because it requires a full daily round trip.

1. Introduction

This project studies the predictability and tradability of close-to-open overnight returns in the U.S. equity market. Unlike close-to-close daily returns, close-to-open returns isolate the price movement from the market close to the next trading day open, a period when firm news, earnings announcements, macroeconomic information, liquidity changes, and investor sentiment may be incorporated into prices while the market is closed.

The objective is to build a machine learning framework that predicts future close-to-open returns and converts those predictions into a dollar-neutral long–short equity portfolio. The problem is formulated as a daily cross-sectional ranking task: stocks with high model scores are selected for the long leg, while stocks with low model scores are selected for the short leg. The strategy is therefore evaluated both statistically, through IC and ablation analysis, and economically, through portfolio performance after liquidity constraints, transaction costs, slippage, and borrow costs.

This distinction is central to the project. A close-to-open strategy has a very short holding period and must trade at both the close and the next open. As a result, even a statistically meaningful signal may fail to generate positive net returns if daily round-trip costs are too large. The central question is therefore not only whether stock-level information can predict close-to-open returns, but whether this predictability remains economically meaningful after realistic implementation frictions.

2. Data and Target Construction

2.1 Data Sources and Panel Structure

The empirical analysis is based on a daily stock level panel constructed from several raw datasets. The core dataset contains daily open, high, low, close, volume, adjustment factors, and market capitalisation for individual stocks. These variables are used to construct adjusted prices, return targets, liquidity measures, volatility estimates, and price-volume features.

Additional datasets provide fundamental scores, GICS industry classifications, earnings information, Piotrosky scores, short-interest variables, and the S&P 500 total return benchmark. The fundamental data provide valuation, quality, health, momentum, and aggregate score variables. The GICS data are used for industry grouping and later cross-sectional processing. Earnings calendar and earnings revision variables capture information related to announcement timing and analyst revisions. Short-interest variables are used both as predictive features and as inputs to a hard-to-borrow proxy. The S&P 500 total return series is retained as the benchmark for later performance evaluation.

All datasets are aligned at the stock-date level. Each row in the final panel represents one stock on one trading date, together with its available characteristics and future return target. Since the prediction horizon is short, careful date alignment is essential. A small timing error can introduce look-ahead bias, especially when features are calculated from same-day closing prices, intraday ranges, or trading volumes.

2.2 Adjusted Prices and Return Target

Before constructing the return target, prices are adjusted for corporate actions. Raw open and close prices may be affected by stock splits, dividends, or other mechanical adjustments. If unadjusted prices were used directly, the target return could contain artificial jumps that do not represent genuine investment returns. The main close-to-open target is therefore computed using adjusted open and adjusted close prices.

The primary prediction target is defined as

$$r^{\mathrm{C2O}}_{i,t+1\to t+2} = \frac{Open^{adj}_{i,t+2}}{Close^{adj}_{i,t+1}} - 1,$$

where i denotes the stock and t denotes the date of the current observation. Under this definition, row t should be interpreted as the prediction row formed at $\tau = (t+1, \text{15:50 ET})$, rather than as a row containing only information from calendar day t. The feature vector $X_{i,t}$ contains information observable no later than 15:50 ET on day $t+1$, and is used to predict the overnight return from the close of day $t+1$ to the open of day $t+2$.

This target differs from the most immediate close-to-open return, $Open_{t+1}/Close_t - 1$, because the strategy forms signals shortly before the closing auction on day $t+1$. At $\tau = (t+1, \text{15:50 ET})$, the model may use information that is observable before the closing auction, but it cannot use the final closing auction price of day $t+1$ or the opening price of day $t+2$. The target is therefore aligned with the tradable close-to-next-open return from day $t+1$ close to day $t+2$ open. This timing convention avoids using information that is only realised after the portfolio formation decision.

This timing convention makes the return target consistent with the actual trading decision: information available before 15:50 ET on day $t+1$ can be used, while the final close of day $t+1$ and the open of day $t+2$ remain unobserved at portfolio formation.

2.3 Earnings Information and Announcement Timing

Earnings information requires separate treatment because the timing of the announcement determines when the information becomes tradable. A before-market announcement and an after-market announcement on the same calendar date do not enter the investor information set at the same time. A mechanical merge by announcement date could therefore assign information to the model before it was actually observable.

The earnings data are aligned using announcement timing. When the before-market or after-market label is missing, the timing is inferred conservatively using a 15:50 cutoff. Earnings features are then shifted to an effective date that reflects when the information could reasonably be used. In addition, an earnings window indicator is constructed so that observations around earnings announcements can be excluded from the tradable set. This reduces exposure to discrete announcement jumps and prevents the strategy from relying on event risk that is difficult to model using daily data.

2.4 Investable Universe Construction

The investable universe is constructed annually using lagged market capitalisation information. For each trading year, stocks are selected based on information available before that year, rather than on future observations. This creates a stable and liquid universe while reducing the risk of look-ahead bias in stock selection.

The yearly universe is then converted into a long table and merged with the main panel. Only stock-date observations that belong to the relevant annual universe are retained. This step ensures that the model and backtest are applied to a consistent set of names that could plausibly have been selected at the start of each trading period.

After constructing the base universe, several diagnostic checks are performed. These include monitoring the number of available stocks through time and comparing cumulative equal-weighted overnight, intraday, and close-to-close returns within the eligible universe. These checks help verify that the universe construction and return alignment are internally consistent.

2.5 Trading Eligibility and Capacity Constraints

The base universe is further refined through trading eligibility filters. These filters determine which stock-date observations are tradable in the portfolio construction stage. The eligibility process includes price, liquidity, volatility, and earnings window restrictions.

Liquidity is particularly important because the feasibility of a trading signal depends on capital scale. A signal that can be implemented at a small portfolio size may not remain tradable at a larger size. The project therefore constructs separate eligibility tables for different portfolio AUM assumptions. As AUM increases, the target dollar position in each stock also increases, making the ADV and participation constraints more restrictive. This allows later results to distinguish between signal quality and capacity limitations.

Table 1 summarises the resulting tradable universe across the three portfolio sizes. The number of eligible names declines as AUM increases. At 50M AUM, the strategy retains around 969 eligible names per day on average. This falls to around 864 names at 250M and 577 names at 1B. The main constraint also changes with capital scale. At 50M and 250M, the annual market cap universe filter remains the largest first binding restriction. At 1B, the ADV liquidity filter becomes the dominant constraint, indicating that capacity rather than raw universe availability is the main limitation at large capital scale.

Table 1: Capacity-aware eligibility summary, 2010–2024

AUM	OK stock-days	Mean/day	Median/day	Main constraint
50M	3,657,102	969.0	977.6	Market-cap universe filter
250M	3,259,409	863.7	898.6	Market-cap universe filter
1B	2,177,155	576.9	565.6	ADV liquidity filter

The eligibility tables also provide the liquidity and volatility variables used for implementation diagnostics. These variables help assess capacity and indicative market impact, while the final portfolio backtest applies the coursework-specified fixed cost schedule.

2.6 Transaction Costs, Slippage, and Borrow Costs

Trading frictions are incorporated before evaluating strategy performance. For the final portfolio backtest, I follow the coursework cost schedule: commission is charged at 1 bp of round-trip traded notional, auction slippage is charged at 3 bps of round-trip traded notional, and borrow costs are charged on the short book.

Separately, I use a square-root market impact framework as a capacity diagnostic. This diagnostic is not the final backtest cost schedule; it is used to show how implementation difficulty varies across stock liquidity, volatility, and AUM. The intuition is that expected price impact increases when the intended trade size is large relative to normal dollar volume, and when the stock is more volatile.

The short side of the portfolio also requires a borrow cost adjustment. Short-interest variables are transformed into a hard-to-borrow proxy through daily cross-sectional ranking. The resulting score assigns stocks into borrow tiers, with each tier mapped to an annualised borrow rate. These annual rates are converted into daily costs and applied to short positions in the backtest.

This treatment is important because the economic value of a close-to-open strategy can differ substantially between gross and net performance. By incorporating commission, the coursework-specified fixed auction-slippage cost, and borrow costs, the final evaluation reflects the practical cost of implementing the signal rather than only its theoretical return.

2.7 Final Modelling Dataset

The final modelling panel combines the adjusted close-to-open target, stock identifiers, dates, market capitalisation, industry labels, engineered features, eligibility indicators, liquidity and cost-related inputs, and borrow cost variables. The feature set includes price-volume signals, return and volatility measures, technical indicators, fundamental scores, earnings revision variables, Piotrosky score, and short-interest measures.

This dataset is designed to support both machine learning prediction and realistic portfolio construction. By enforcing a clear trading timeline, using adjusted prices, constructing the universe with lagged information, and incorporating capacity and cost variables, the data construction process provides a reliable foundation for evaluating whether the close-to-open signal is economically meaningful rather than only statistically detectable.

3. Tradability and Capacity Filters

3.1 Thresholds and Numerical Justification

Section 2 establishes the stock-level research panel by constructing the delayed close-to-open return target and defining the annual investable universe. The baseline universe consists of the prior-year-end top 1,000 stocks by market capitalisation. By fixing the eligible names before each trading year begins, the design avoids using within-year market-cap information during portfolio formation and therefore reduces the risk of look-ahead bias.

This section moves from data construction to implementation feasibility. It extends the analysis by translating the eligible universe into a tradable universe, applying practical filters, estimating transaction-cost effects, and assessing how these choices affect portfolio capacity across different AUM levels. It also explains which constraints drive stock-day exclusions, helping to interpret the strategy's capacity limits.

The market capitalisation screen builds on Section 2, where the investable universe is defined using a market-capitalisation ranking rather than a fixed market-capitalisation threshold. Specifically, for each trading year, the sample is restricted to the top 1,000 stocks by prior-year-end market capitalisation.

Although the annual universe contains only 1,000 stocks, the price per share filters are applied at the stock-day level. Over 2010–2024, this produces massive stock-day observations. Within this universe, most stock-days are expected to have prices above \$5. A stock-day is classified as tradable only when its closing price satisfies

$$\mathrm{Close}_{i,t} > \$5.$$

The \$5 cutoff is used because, below this level, a one-cent tick represents at least 20 basis points of price movement, making percentage spreads, price discreteness, and closing-auction execution noise more important.

After applying the annual market-cap universe and the price screen, 18,652 stock-days are excluded over 2010–2024 because their closing price is less than or equal to \$5.

The dollar-volume filter is based on average daily dollar volume, where daily dollar volume is measured as closing price multiplied by trading volume and then averaged over the previous 20 trading days:

$$ADV20_i = \frac{1}{20} \sum_{s=t-19}^{t} Close_{i,s} \times Volume_{i,s}.$$

The portfolio is constructed with 200 long and 200 short positions, so each stock receives a target dollar position of:

$$Position_i = \frac{AUM}{200 + 200} = \frac{AUM}{400}.$$

The capacity rule requires the stock's ADV20 to be at least twenty times the target position:

$$ADV20_i > 20 \times Position_i = 20 \times \frac{AUM}{400}.$$

Equivalently, the implied participation rate is capped at 5%:

$$Participation_i = \frac{Position_i}{ADV20_i} < \frac{1}{20} = 5\%.$$

Table 2: Dollar-volume capacity thresholds by AUM.

AUM	ADV threshold	Participation cap	Per-stock target
50M	\$2.5m	5.0%	\$0.125m
250M	\$12.5m	5.0%	\$0.625m
1B	\$50.0m	5.0%	\$2.500m

Larger AUM requires a proportionally larger ADV20 threshold. Therefore, the same signal can remain tradable at smaller AUM but become capacity constrained at larger AUM because fewer stocks have enough dollar volume to support the required position size.

The realized-volatility filter uses a 252-trading-day close-to-close volatility estimate from adjusted close returns and annualizes it by $\sqrt{252}$. The accepted range is $10\% < \sigma_{ann} < 200\%$. The lower bound removes stale or mechanically flat price series where return estimates are unreliable; the upper bound removes distressed or event-driven observations where square-root impact and daily close-to-open signals are unlikely to be stable.

The earnings filter is a one-effective-trading-date blackout. Missing before/after-market labels are inferred using a 15:50 cutoff. Before-market announcements are assigned to the previous trading date; after-market announcements remain on the reporting date. The excluded observation is the signal row whose next implemented trading date equals the effective earnings date. The resulting blackout is a same-effective-date exclusion rather than a multi-day earnings window.

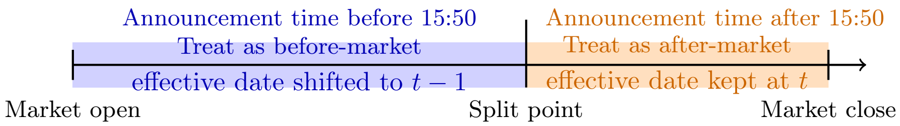


Figure 1: Earnings announcement timing rule based on the 15:50 cutoff.

3.2 Closing-Auction Participation and Slippage

The closing-auction participation cap is derived from the liquidity requirement rather than introduced as a separate assumption. Each stock receives an equal target dollar position equal to AUM divided by 400, reflecting the 200 long and 200 short portfolio construction. The ADV20 screen requires trailing 20-day average dollar volume to exceed 20 times this target position. This is equivalent to requiring the intended trade size to be no more than one twentieth of ADV20. The implied closing-auction participation cap is therefore 5%.

As an implementation diagnostic, I estimate indicative auction impact using the square-root impact model:

$$\text{impact bps} = 0.7 \times \sigma_{\text{daily}} \times \sqrt{f} \times 10{,}000,$$

where $f = 5\%$ is the assumed participation rate and σ_{daily} is daily realised volatility. At the 5% ADV participation cap, the typical large-cap stock-day is defined as market-cap ranks 1 to 250 within the annual top-1000 universe, while the typical mid-cap stock-day is defined as ranks 751 to 1000. These definitions use median stock-day characteristics within each bucket rather than selecting a single named stock.

Table 3: Diagnostic square-root auction impact at the 5% ADV participation cap

Bucket	Stock-days	ADV20	Ann. vol	Daily vol	Impact	Impact + comm.
Large-cap ranks 1–250	943,498	\$299.5m	24.7%	1.56%	24.40 bps	24.90 bps
Mid-cap ranks 751–1000	943,488	\$17.6m	33.3%	2.10%	32.81 bps	33.31 bps

These square-root estimates are not used as the final cost schedule in the portfolio backtest. They are reported only to justify the capacity screen and to show why larger AUM and lower-liquidity stocks are more difficult to trade. The final headline performance uses the coursework-specified fixed slippage cost of 3 bps of round-trip traded notional.

The results show that the same 5% participation cap implies different implementation costs across the universe. The large-cap bucket has much higher median ADV20 and lower realised volatility, so its estimated auction impact is 24.40 bps before commission. The mid-cap bucket has lower trading capacity and higher realised volatility, so its estimated impact rises to 32.81 bps. After adding the 0.5 bp commission component, the estimated total trading cost is 24.90 bps for the large-cap bucket and 33.31 bps for the mid-cap bucket. This provides a numerical justification for treating liquidity capacity as a binding portfolio-formation constraint.

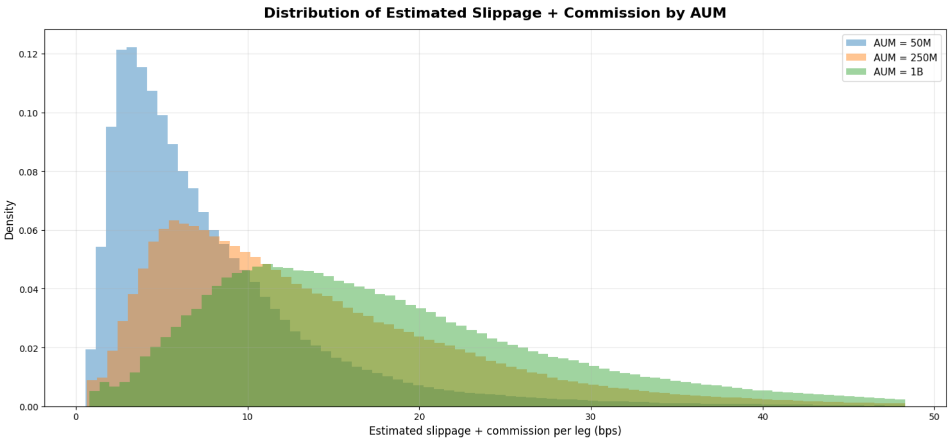


Figure 2: Distribution of estimated slippage plus commission across stock-days at different AUM levels.

Figure 2 complements the representative slippage estimates by showing the full distribution of estimated implementation costs across stock-days and AUM levels. As AUM increases from \$50 million to \$250 million and \$1 billion, the distribution shifts progressively to the right. This reflects the fact that the target dollar position per stock increases with portfolio size, raising the trade size relative to ADV20. Under the square-root impact model, expected trading cost increases with the square root of the participation rate. Therefore, larger portfolios face higher implementation costs even after the liquidity screen is applied. The \$1 billion portfolio exhibits both a higher central cost level and a longer right tail, indicating that closing-auction participation becomes a materially binding capacity constraint at larger scale.

3.3 Eligible-Set Evolution, 2010–2024

After applying filters, the eligible set varies strongly with portfolio AUM. The 50M portfolio remains close to the full universe throughout the sample, rising from 914.7 average eligible names per day in 2010 to 982.5 in 2024. The 250M portfolio is more capacity-constrained early in the sample, with 667.5 average eligible names per day in 2010, but it converges toward the 50M portfolio by 2024. The 1B portfolio is materially more constrained, increasing from 411.9 eligible names per day in 2010 to 777.1 in 2024, but remaining below the smaller-AUM portfolios throughout the sample.

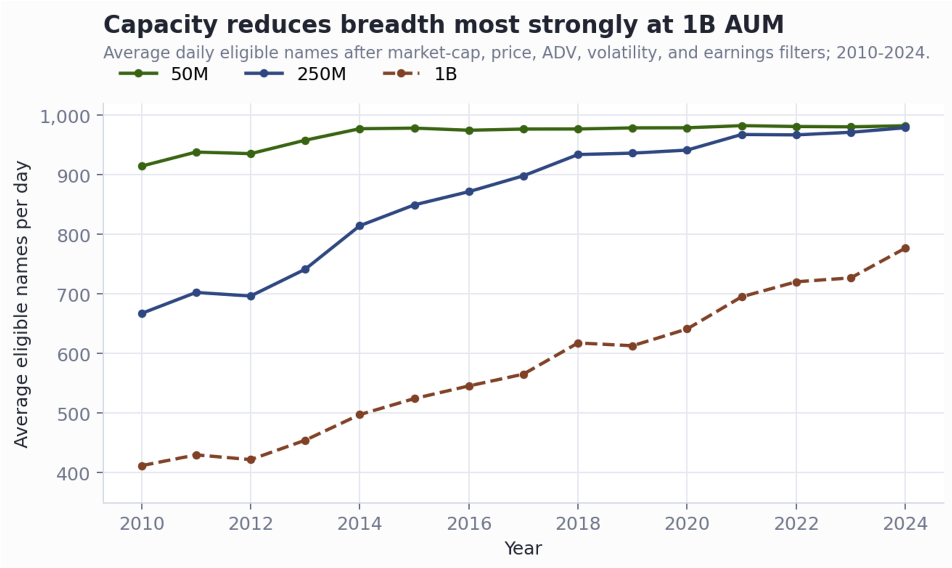


Figure 3: Eligible-set evolution by portfolio AUM, 2010–2024.

Figure 3 shows that capacity becomes less restrictive over time, particularly for the 250M and 1B portfolios. This is consistent with the growth in market liquidity over the sample period. However, the separation between the three lines also shows that the ADV screen remains economically meaningful: increasing AUM directly increases the required dollar volume per stock, so larger portfolios lose more names even within the same top-1000 universe.

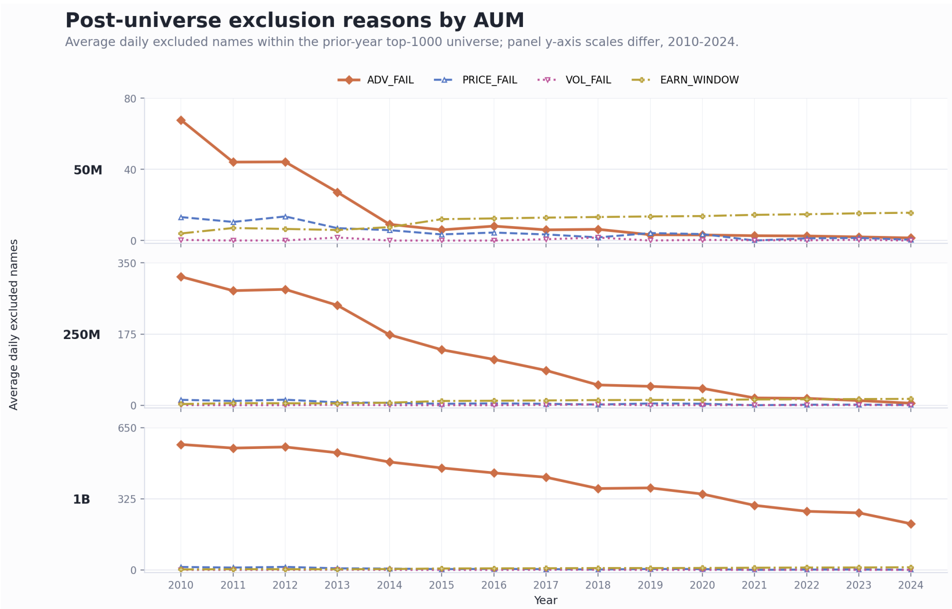


Figure 4: Post-universe exclusion reasons by AUM, 2010–2024.

The y-axis scale differs by AUM panel to make within-portfolio trends visible. The dominant post-universe exclusion is `ADV_FAIL`, which declines over time for all three AUM levels. This indicates that liquidity capacity improves through the sample, while price, volatility, and earnings-window exclusions remain much smaller.

3.4 Binding-Constraint Distribution

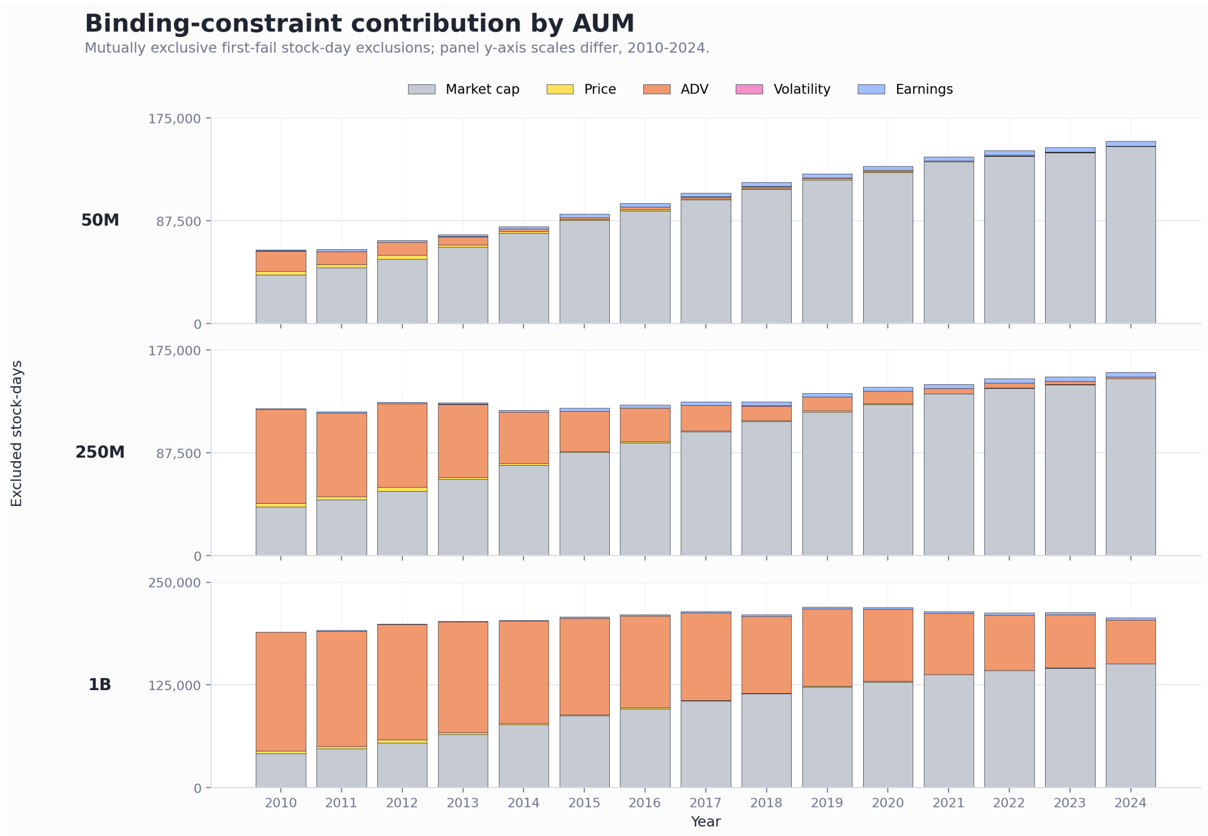


Figure 5: Binding-constraint contribution by AUM, 2010–2024.

The market-cap component reflects the annual restriction to the prior-year top-1000 universe, so it is mainly a universe-definition screen. The main capacity-sensitive constraint is `ADV_FAIL`.

Figure 6 checks the COVID window separately. Using 2019H2 as the pre-shock baseline, 2020H1 as the stress period,the volatility filter becomes visible in early 2020 but remains small in magnitude. The maximum monthly share of stock-days outside the 10%–200% annualised volatility band is about 0.21%. Thus, the main binding constraint during this period is still liquidity capacity rather than volatility exclusion.

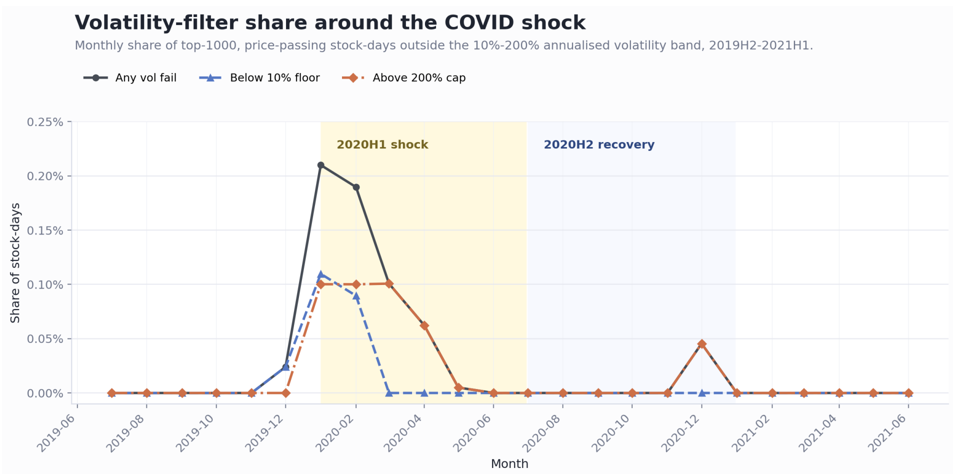


Figure 6: Volatility-filter share around the COVID shock, 2019H2–2021H1.

4. Borrow and Hard-to-Borrow Treatment

4.1 Hard-to-Borrow Proxy

The hard-to-borrow proxy is estimated from the delayed short-interest feature file. Each short-interest observation is shifted forward by 10 calendar days, matching the public-reporting lag rather than allowing unreleased short-interest information into the trading decision. The most recently available short-interest observation is then carried forward for each instrument.

The proxy inputs are:

- dsi: short-interest intensity.
- $dtcn$: days to cover.
- $ddtcn$: change in days to cover.

For each trading date, each input is ranked cross-sectionally on a percentile scale and missing ranks are set to zero. The combined score is:

$$HTB = 0.50 \times rank(dsi) + 0.30 \times rank(dtcn) + 0.20 \times rank(ddtcn).$$

Borrow tier A is the default tier. Tier B begins at $HTB \geq 0.80$, tier C begins at $HTB \geq 0.95$, and an override also forces tier C when $rank(dsi) \geq 0.95$ and $rank(dtcn) \geq 0.90$. The annual borrow costs are 40 bps, 200 bps, and 800 bps for tiers A, B, and C respectively, converted to daily costs by dividing by 252.

Table 4: Hard-to-borrow tier distribution in the top-1000 universe.

Borrow tier	Stock-days	Fraction
A	3,292,240	87.24%
B	369,930	9.80%
C	111,801	2.96%

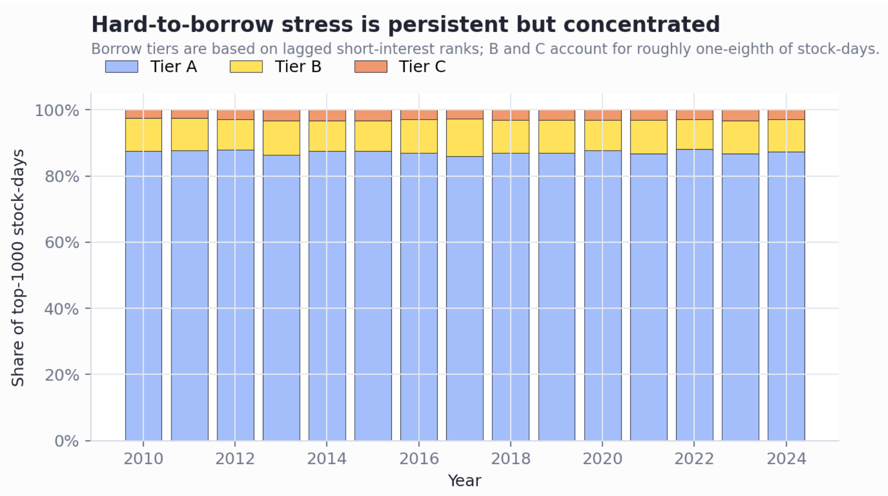


Figure 7: Annual hard-to-borrow tier shares in the top-1000 universe.

Because stock-level borrow-fee data are not available, the hard-to-borrow proxy is converted into a stylised borrow-cost schedule. Tier A is assigned 40 bps per year, representing general-collateral borrowing. Tier B is assigned 200 bps per year, representing moderately hard-to-borrow names. Tier C is assigned 800 bps per year, representing crowded or hard-to-borrow shorts. The schedule is therefore an implementation assumption used to quantify sensitivity to borrow frictions.

4.2 External Validation

Since real-time prime-broker hard-to-borrow data are not available, the validation is based on whether the proxy identifies stocks that were later documented as crowded and expensive-to-borrow shorts. The short-interest inputs are treated as lagged information, consistent with FINRA's twice-monthly short-interest reporting schedule.[1]

Figure 8 compares the daily point-in-time HTB score with external short-squeeze evidence for GameStop (GME) and Macerich (MAC). For GME, the SEC staff report states that lending fees were around 25% in January 2021, while Securities Finance Times reported an average cost-to-borrow of about 33% on 28 January 2021.[2] The proxy places GME in tier B or C on 51.3% of stock-days from July 2020 to March 2021. For MAC, external reports identify short interest of 58% at 31 December 2020 and describe the stock as part of the January 2021 WallStreetBets short-squeeze episode.[3] The proxy places MAC in tier B or C on 73.5% of stock-days over the same validation window.

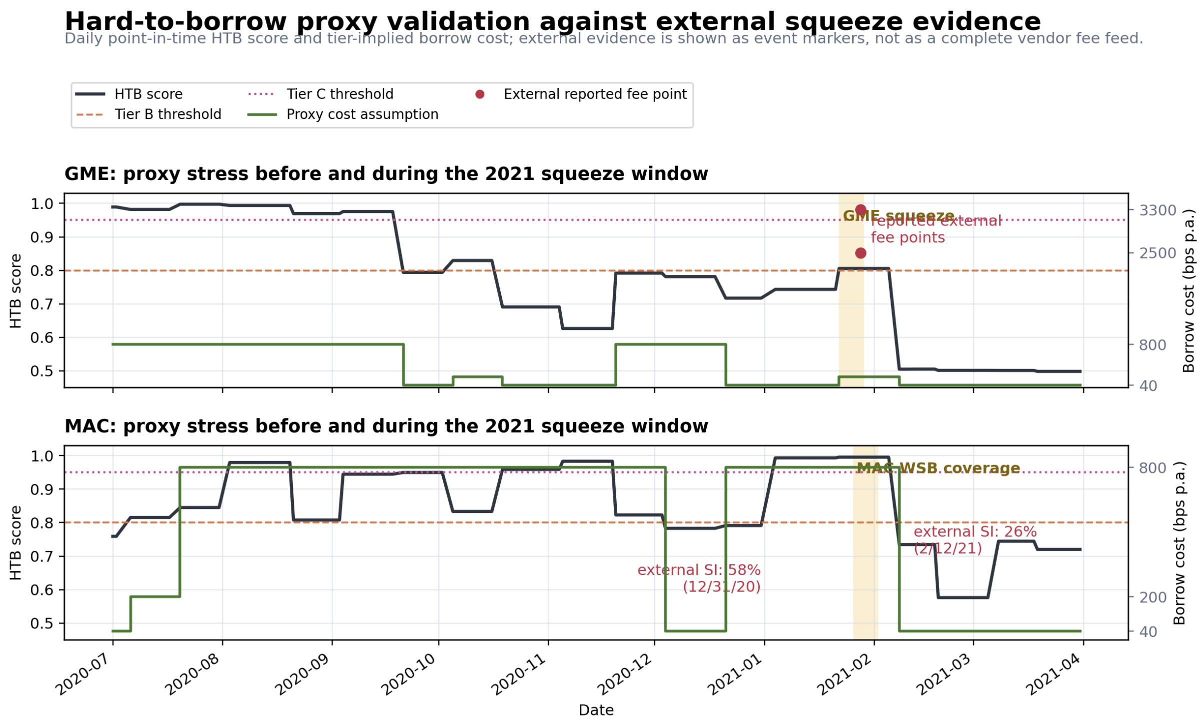


Figure 8: Hard-to-borrow proxy validation against external short-squeeze evidence.

This validation is directional rather than a full borrow-fee calibration. The proxy correctly flags both names as crowded short-risk stocks before or during the documented squeeze window. However, the tiered cost schedule caps tier C at 800 bps per year, so it deliberately understates extreme cases such as GME, where reported borrow fees reached much higher levels.

4.3 Exclusion Versus Tiered Cost

The adopted treatment is tiered cost only. No stock-day is hard-excluded solely because it is expensive to borrow. Instead, the daily borrow rate is charged to short positions according to the A/B/C tier. This preserves the raw short signal but makes crowded shorts pay for borrow stress.

[1] FINRA short-interest reporting: https://www.finra.org/filing-reporting/regulatory-filing-systems/short-interest.

[2] SEC Staff Report on Equity and Options Market Structure Conditions in Early 2021: https://www.sec.gov/file/staff-report-equity-options-market-struction-conditions-early-2021pdf. Securities Finance Times, "GameStop: The real data story and what it means for securities lending": https://www.securitiesfinancetimes.com/securitieslendingnews/industryarticle.php?article_id=224507.

[3] Motley Fool, "1 Real Estate Stock With Even Higher Short Interest Than GameStop," 25 February 2021: https://www.fool.com/investing/2021/02/25/1-real-estate-stock-with-even-higher-short-interes/.

Table 5: Fraction of stock-days affected by the tiered borrow treatment.

Measure	Fraction
Universe stock-days in B or C	12.76%
250M eligible stock-days in B or C	12.09%

The fraction in Table 5 should be interpreted as exposure to the borrow-cost schedule. The first row is the top-1000 universe exposure to the borrow treatment. The second row uses the 250M post-capacity eligible set as a proxy for the final shortable pool. The selected-short exposure is evaluated through the portfolio-level Sharpe decomposition below, while this table reports the broader universe and eligible-set exposure to the borrow-cost schedule.

4.4 Gross-of-Borrow and Net-of-Borrow Sharpe

The borrow-cost impact is measured after applying the non-borrow trading costs from the Section 6 cost schedule. Therefore, "gross-of-borrow" refers to the Sharpe ratio after commission and slippage but before borrow costs. "Net-of-borrow" refers to the Sharpe ratio after commission, slippage, and the tiered borrow-cost schedule.

Table 6: Gross-of-borrow and net-of-borrow Sharpe ratios.

AUM	Gross-of-borrow Sharpe	Net-of-borrow Sharpe	Borrow Sharpe drag
50M	-1.42	-1.53	0.11
250M	-1.41	-1.52	0.11
1B	-1.20	-1.30	0.10

The tiered borrow treatment reduces Sharpe by about 0.10–0.11 points across the three AUM levels. This confirms that hard-to-borrow costs matter for short-book realism, but they are not the main source of performance degradation. The strategy is already negative before borrow costs are applied, because daily close-to-open trading makes commission and auction slippage the dominant cost components.

5. Constructing the cross-sectional alpha

5.1 Information Set and Prediction Target

The portfolio formation time is $\tau = (t+1, 15{:}50 \text{ ET})$. At this time, the model forms positions that enter at the closing auction of day $t+1$ and exit at the opening auction of day $t+2$. The information set is denoted by $\mathcal{F}_\tau$, and contains only information observable no later than 15:50 ET on day $t+1$.

The prediction target is the next close-to-open return:

$$r^{\text{C2O}}_{i,t+1\to t+2} = \frac{O^{\text{adj}}_{i,t+2}}{C^{\text{adj}}_{i,t+1}} - 1.$$

Here $C^{\text{adj}}_{i,t+1}$ is the adjusted close used for the closing-auction entry on day $t+1$, and $O^{\text{adj}}_{i,t+2}$ is the adjusted open used for the opening-auction exit on day $t+2$. This return is not observable at τ, so it is outside the information set: $r^{\text{C2O}}_{i,t+1\to t+2} \notin \mathcal{F}_\tau$.

The model is trained as a cross-sectional classifier. On each training date, stocks in the top p fraction of realised $r^{\text{C2O}}_{i,t+1\to t+2}$ are assigned label $+1$, stocks in the bottom p fraction are assigned label -1, and middle observations are excluded from classifier training.

The model output is a real-valued score $s_{i,t}$. A higher score means a higher expected close-to-next-open return.

5.2 Feature Definitions and Observability

All model inputs are constructed to be measurable with respect to $\mathcal{F}_\tau$, meaning that they must be observable no later than 15:50 ET on day $t+1$. The final feature set contains approximately 250 variables, including market capitalisation controls, and is organised into five groups:

- **Fundamental and valuation**: valuation, quality, health, momentum, composite fundamental scores, value-trap indicators, and market-cap-related variables.
- **Earnings and analyst revisions**: earnings surprises, EPS variables, analyst revision proxies, and smoothed revision changes.
- **Returns, volatility, and liquidity**: lagged returns, overnight returns, realised volatility, turnover, volume surprise, and price-channel variables.
- **Technical indicators**: OHLCV-based trend, moving-average, slope, Hilbert-transform, and high-low-close indicators.
- **Short interest and borrow-related**: short-interest ratio, days-to-cover, changes in short-interest pressure, and borrow-related signals.

These five groups are used both for model training and for the leave-one-group-out ablation analysis.

5.3 Model Class and Training Scheme

The final model is a random forest classifier trained as a daily cross-sectional tail classifier. Each dataframe row indexed by t contains a feature vector $X_{i,t}$ that is observable by the portfolio formation time $\tau = (t+1, 15{:}50\text{ ET})$. Here, t is the prediction-row index used to align features with the future close-to-open target; it does not imply that the feature vector is restricted to information available only at the end of calendar day t. The target attached to row t is the future close-to-next-open return:

$$r^{\mathrm{C2O}}_{i,t+1\to t+2} = \frac{O^{\mathrm{adj}}_{i,t+2}}{C^{\mathrm{adj}}_{i,t+1}} - 1.$$

In code, this corresponds to $\texttt{open}_{i,t+2}/\texttt{close}_{i,t+1} - 1$.

The classifier is trained only on the cross-sectional tails of the realised future return. With `label_pct` $= 0.20$, the top 20% of stocks receive label $+1$, the bottom 20% receive label -1, and the middle 60% are excluded from classifier fitting:

$$y_{i,t} = \begin{cases} +1, & r^{\mathrm{C2O}}_{i,t+1\to t+2} \ge q_{0.80,t}, \\ -1, & r^{\mathrm{C2O}}_{i,t+1\to t+2} \le q_{0.20,t}. \end{cases}$$

The model score is the probability spread between the positive and negative classes:

$$s_{i,t} = \Pr(y_{i,t} = +1 \mid X_{i,t}) - \Pr(y_{i,t} = -1 \mid X_{i,t}).$$

Higher scores indicate stronger long candidates, while lower scores indicate stronger short candidates.

The model uses a rolling point-in-time training scheme with `whole_train_days` $= 126$, `val_days` $= 21$, and `predict_days` $= 21$. For each prediction block, the first 105 trading days are used for training and feature filtering, the last 21 trading days are used for validation, and the selected model is then refit on the full 126-day window before predicting the next 21 trading days.

Before hyperparameter tuning, features are grouped by absolute Spearman correlation within the training split. The threshold is $|\rho_{\text{Spearman}}| > 0.75$. A feature can join a cluster only if its absolute Spearman correlation exceeds 0.75 with every feature already in that cluster. A preliminary random forest is trained on the training split, and cluster-level permutation importance is evaluated on the validation split by shuffling all features in the same cluster together. Clusters with negative validation permutation importance are removed.

The retained feature clusters are compressed using one principal component per cluster:

$$Z_{g,t} = \mathrm{PC1}(X_{g,t}),$$

where g denotes a retained feature cluster. During validation, PCA is fit only on the training split and applied to the validation split. After hyperparameter selection, PCA is refit on the full 126-day training-validation window and applied to the out-of-sample prediction block.

The random forest uses `criterion=log_loss`, `bootstrap=True`,

and `class_weight=balanced_subsample`.

The hyperparameter grid is:

$$\texttt{n\_estimators} \in \{100\}, \texttt{max\_depth} \in \{1,2\}, \texttt{min\_samples\_leaf} \in \{100\}, \texttt{max\_features} \in \{\sqrt{p}, 0.2\}.$$

Hyperparameters are selected using validation portfolio return, `selection_metric=return`. For each candidate parameter set, the model is trained on the 105-day training split, scored on the 21-day validation split, and evaluated by the validation long-short portfolio return.

This model class is appropriate because the problem is daily, cross-sectional, nonlinear, and noisy. Random forests can capture nonlinear interactions across fundamentals, earnings revisions, technical indicators, liquidity variables, and short-interest variables while remaining relatively robust in short rolling windows. The rolling scheme avoids look-ahead bias because feature filtering, cluster selection, PCA fitting, hyperparameter selection, final refitting, and prediction are all performed using only data available before the out-of-sample prediction block.

5.4 Cross-Sectional Information Coefficient

I evaluate the model using the daily cross-sectional Spearman Information Coefficient (IC). For each date t, the IC is computed as the Spearman rank correlation between the model score $s_{i,t}$ and the realized close-to-next-open return:

$$\mathrm{IC}_t = \mathrm{corr}_i^{\mathrm{Spearman}}\left(s_{i,t}, r^{\mathrm{C2O}}_{i,t+1\to t+2}\right).$$

The reported IC mean is the time-series average of daily ICs, and the IC t-statistic is

$$t_{\mathrm{IC}} = \frac{\overline{\mathrm{IC}}}{\sigma(\mathrm{IC})}\sqrt{N},$$

where N is the number of daily IC observations.

For the random forest model, the overall cross-sectional IC is positive and statistically significant:

$$\overline{\mathrm{IC}} = 0.023179, \quad \sigma(\mathrm{IC}) = 0.148463, \quad t_{\mathrm{IC}} = 6.062745, \quad N = 1508.$$

This indicates that the model has a persistent positive cross-sectional ranking ability over the out-of-sample period.

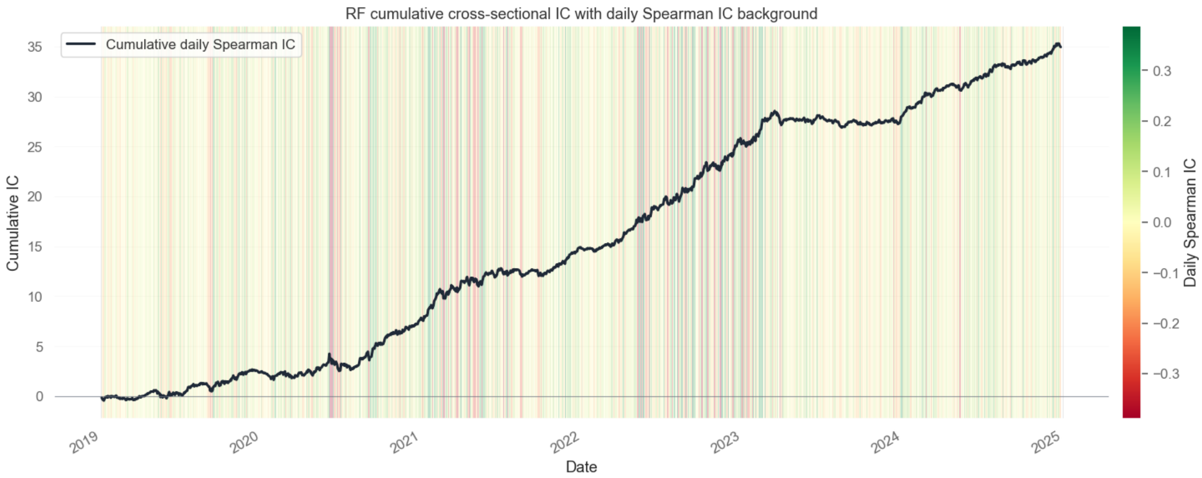


Figure 9: Cumulative daily cross-sectional IC for the random forest model.

Stability across years. The yearly IC results are reported below:

Year	IC Mean	IC Std	t-stat	N
2019	0.009707	0.104589	1.473405	252
2020	0.020681	0.151313	2.173965	253
2021	0.028208	0.169504	2.641781	252
2022	0.041592	0.190210	3.464275	251
2023	0.010944	0.140060	1.235526	250
2024	0.027963	0.116473	3.795975	250

The IC is positive in every year from 2019 to 2024. The strongest annual IC occurs in 2022, with an IC mean of 0.041592 and a t-statistic of 3.464275. The weakest years are 2019 and 2023, but both still have positive average ICs.

Stability across market regimes. I define market regimes using the equal-weighted cross-sectional average return on each date. Dates with sufficiently positive average returns are classified as up markets, dates with sufficiently negative average returns are classified as down markets, and the remaining dates are classified as flat markets.

The IC by regime is:

Regime	IC Mean	IC Std	t-stat	N
down_market	0.033389	0.171048	4.294493	484
flat_market	0.025781	0.107534	4.655111	377
up_market	0.014024	0.150489	2.370363	647

The model has positive IC in all three regimes. The ranking signal is strongest in down markets and flat markets, with IC means of 0.033389 and 0.025781, respectively. The IC remains positive in up markets, but is weaker at 0.014024.

Year-regime stability. The model is not uniformly strong in every sub-period. Some year-regime combinations show weak or negative IC. For example, the model performs poorly in 2021 down markets:

$$\overline{\mathrm{IC}} = -0.152596, \quad t_{\mathrm{IC}} = -7.730709, \quad N = 57.$$

It also performs poorly in 2022 up markets:

$$\overline{\mathrm{IC}} = -0.100328, \quad t_{\mathrm{IC}} = -5.944885, \quad N = 95.$$

These regime-specific weaknesses suggest that the model's ranking signal is sensitive to the direction and structure of the market environment. In particular, the model is strongest when the market regime is more defensive or flat, and weaker during some strong up-market periods where broad beta or momentum effects may dominate stock-specific cross-sectional signals.

Overall, the full-period IC is positive and statistically significant, the yearly IC is positive in every year, and the regime analysis shows that the signal is robust on average but not uniformly stable across all market environments.

5.5 Feature Group Ablation

I evaluate feature contribution using a leave-one-group-out ablation. For each ablation, one feature group is removed, the full random forest pipeline is retrained from scratch, and out-of-sample predictions are regenerated. The same rolling training, feature clustering, cluster permutation filtering, PCA compression, validation-based hyperparameter selection, and final refitting procedure is repeated for every ablation.

This ablation is mainly a signal-quality diagnostic. The Sharpe ratios below are computed from a simplified 50% long and 50% short portfolio based directly on model scores. They do not include the final eligibility filters, participation-cap sizing, transaction costs, auction slippage, or borrow costs. Therefore, Sharpe is interpreted as a portfolio-level diagnostic, while IC is the cleaner measure of pure cross-sectional prediction ability.

The full model has a diagnostic Sharpe of 1.351041. I define $\Delta\text{Sharpe} = \text{Sharpe}_{\text{ablation}} - \text{Sharpe}_{\text{full}}$.

Removed Feature Group	Sharpe	ΔSharpe	Ann. Return	ΔAnn. Return
technical_indicator	1.575946	0.224905	0.029454	0.003567
short_interest	1.419488	0.068448	0.026944	0.001056
fundamental	1.385579	0.034538	0.026830	0.000943
earnings	1.309208	−0.041833	0.024676	−0.001212
returns_volatility_liquidity	1.206386	−0.144655	0.021084	−0.004803

The simplified Sharpe ablation suggests that removing the returns, volatility, and liquidity group hurts portfolio performance the most, reducing Sharpe from 1.351041 to 1.206386. Removing earnings also mildly reduces Sharpe to 1.309208. By contrast, removing technical indicators or short-interest variables improves the simplified diagnostic Sharpe, which suggests that these groups may affect tail portfolio construction differently from their effect on overall ranking quality.

To focus more directly on prediction ability, I also evaluate IC for each ablation model:

Model	IC Mean	IC Std	t-stat	N
rf_full	0.023179	0.148463	6.062745	1508
rf_without_earnings	0.022351	0.147015	5.903979	1508
rf_without_fundamental	0.022063	0.150779	5.682320	1508
rf_without_returns_volatility_liquidity	0.019707	0.136361	5.612323	1508
rf_without_short_interest	0.023094	0.146662	6.114841	1508
rf_without_technical_indicator	0.019723	0.141379	5.417372	1508

The IC results give a clearer view of predictive contribution. The full model has an IC mean of 0.023179 with a t-statistic of 6.062745. Removing technical indicators lowers IC to 0.019723, while removing returns, volatility, and liquidity lowers IC to 0.019707. These are the two largest IC declines, indicating that technical indicators and return-volatility-liquidity variables are the most important groups for cross-sectional ranking.

The difference between Sharpe and IC is informative. Technical indicators improve overall ranking ability, but their removal improves the simplified 50%/50% diagnostic Sharpe. A plausible explanation is that technical indicators improve broad cross-sectional ordering while also increasing exposure to noisy or high-volatility tail names in the simplified portfolio. Since this diagnostic portfolio ignores eligibility, capacity, transaction costs, and borrow costs, IC is the more reliable measure of pure model prediction quality here.

Short-interest variables contribute little to IC: removing them changes IC only from 0.023179 to 0.023094. Fundamentals and earnings provide smaller positive contributions, with IC falling to 0.022063 and 0.022351, respectively. Importantly, no single feature-group removal collapses the model. Even after removing the two most important IC groups, the ablated models retain IC means around 0.0197 with t-statistics above 5. This suggests that the signal is diversified across multiple feature families rather than depending on one fragile source of alpha.

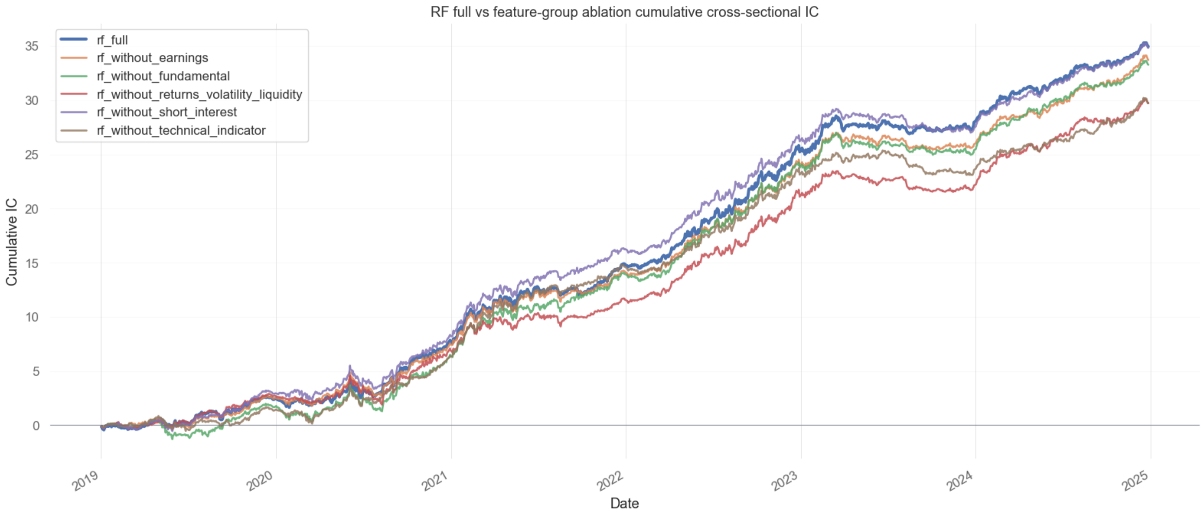


Figure 10: Cumulative cross-sectional IC for the full random forest and leave-one-feature-group-out ablation models. This ablation is evaluated as a signal-quality diagnostic before final eligibility filtering, participation-cap sizing, transaction costs, auction slippage, and borrow costs are applied.

Overall, the ablation shows that technical indicators and return-volatility-liquidity variables contribute the most to the model's cross-sectional ranking ability, as measured by IC. Earnings and fundamental variables provide smaller incremental contributions, while short-interest variables have limited alpha contribution in the model itself. The simplified Sharpe ablation is useful as a portfolio-level diagnostic, but because it ignores final eligibility and cost constraints, the IC ablation is the more reliable measure of predictive contribution in this section.

5.6 Weak Regimes and Model Limitations

Although the full-period IC is positive and significant,

$$\overline{\mathrm{IC}} = 0.023179, \qquad t_{\mathrm{IC}} = 6.062745,$$

the model is not uniformly strong across all regimes.

The weakest regime is 2021 down markets. In this sub-period, the IC is strongly negative:

$$\overline{\mathrm{IC}} = -0.152596, \qquad t_{\mathrm{IC}} = -7.730709, \qquad N = 57.$$

This suggests that the model ranked stocks in the wrong direction during those sell-off days. A likely explanation is that down markets in 2021 were driven by sharp rotations and risk-off moves, where factor relationships changed quickly and defensive or crowded names behaved differently from the recent training window.

Another weak regime is 2022 up markets:

$$\overline{\mathrm{IC}} = -0.100328, \qquad t_{\mathrm{IC}} = -5.944885, \qquad N = 95.$$

During this period, broad market rebounds and macro-driven moves likely dominated stock-specific signals.

The model is trained on cross-sectional features and short rolling windows, so it may underperform when returns are driven more by common beta, sector rotation, or short-covering than by the firm-level signals in the feature set.

The model also shows weaker performance in the full-year 2019 and 2023 samples:

$$\overline{\mathrm{IC}}_{2019} = 0.009707, \qquad t_{\mathrm{IC},2019} = 1.473405,$$

$$\overline{\mathrm{IC}}_{2023} = 0.010944, \qquad t_{\mathrm{IC},2023} = 1.235526.$$

These years still have positive IC, but the signal is weaker and less statistically significant than in 2022 or 2024.

Overall, the model performs badly when market behavior is dominated by regime shifts, sharp rotations, or broad common-factor moves. In those environments, the learned cross-sectional relationships from the recent rolling window are less stable, and the model's stock ranking ability deteriorates.

6. From Ranking to Portfolio, with Realistic Costs

6.1 Basket Construction, Weighting, and Turnover

The portfolio construction is AUM-aware. In the liquidity and ADV eligibility checks, I assume a maximum long-short book of 200 long names and 200 short names. Therefore, the target notional of each selected single-stock position is

$$\text{Position per stock} = \frac{\text{AUM}}{400}.$$

For the three AUM levels, this gives:

AUM	Portfolio AUM	Position per Stock
50M	50,000,000	125,000
250M	250,000,000	625,000
1B	1,000,000,000	2,500,000

The trading signal is based on the model score. I use a zero-centered threshold rule with one cross-sectional standard deviation:

$$\texttt{score\_center\_method} = \texttt{zero}, \qquad \texttt{score\_std\_multiplier} = 1.$$

Thus, on each date, the long and short thresholds are

$$\theta_t^L = \sigma_t(s), \qquad \theta_t^S = -\sigma_t(s),$$

where $\sigma_t(s)$ is the cross-sectional standard deviation of model scores on date t.

Only eligible stocks are allowed to enter the basket. The candidate sets are

$$L_t^{\text{cand}} = \{i : s_{i,t} \geq \theta_t^L,\ i \text{ is eligible}\},$$

$$S_t^{\text{cand}} = \{i : s_{i,t} \leq \theta_t^S,\ i \text{ is eligible}\}.$$

To keep the portfolio dollar-neutral, I use the smaller side count:

$$N_t = \min\left(|L_t^{\text{cand}}|, |S_t^{\text{cand}}|, 200\right).$$

The long basket contains the top N_t stocks by score, and the short basket contains the bottom N_t stocks by score. Because each stock receives the same dollar notional, the long and short dollar exposures are equal:

$$\text{Long Notional}_t = N_t \frac{\text{AUM}}{400}, \qquad \text{Short Notional}_t = N_t \frac{\text{AUM}}{400}.$$

The realised basket sizes and turnover are:

AUM	Avg Long	Avg Short	Gross Exp.	Long Exp.	Short Exp.	Avg Turnover
50M	127.09	127.09	0.635468	0.317734	0.317734	0.635468
250M	124.14	124.14	0.620723	0.310361	0.310361	0.620723
1B	84.46	84.46	0.422304	0.211152	0.211152	0.422304

The average basket size falls as AUM increases because the eligibility filter becomes stricter at larger portfolio sizes. The 1B portfolio therefore uses materially lower gross exposure than the 50M and 250M portfolios.

Turnover is especially important in this strategy. Each position is a close-to-next-open trade: the portfolio enters near the close and exits at the next open. Therefore, even if the same stock appears in the basket on two consecutive days, I still treat it as a new daily trade. I do not net today's position against yesterday's position when computing turnover.

Daily turnover is defined as

$$\text{Turnover}_t = \frac{\text{Daily Traded Dollars}_t}{\text{AUM}}.$$

Since every selected stock is traded each day,

$$\text{Daily Traded Dollars}_t = 2N_t \frac{\text{AUM}}{400}.$$

Thus, the average daily turnover is approximately equal to the average gross exposure. I use the zero plus/minus one standard deviation threshold because I want to reduce turnover relative to trading the entire cross-section, while still keeping enough names for diversification. However, the strategy still pays both sides of transaction costs every day because the holding period is close-to-next-open. This high daily turnover causes transaction costs and slippage to severely erode the gross alpha.

6.2 Headline Performance at Different AUM Levels

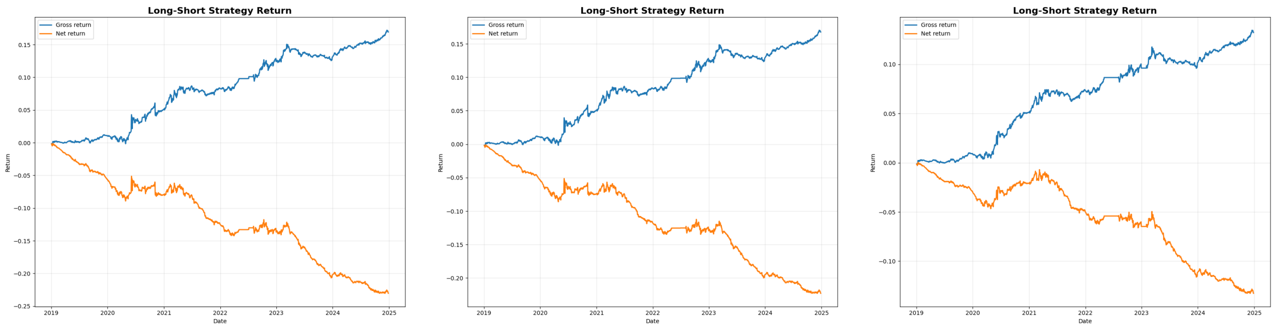
I report the headline performance at three portfolio-AUM levels: 50M, 250M, 1B.

The predictive out-of-sample evaluation starts in 2019 because the earlier sample is used to build sufficient point-in-time feature history, rolling transformations, normalization windows, and rolling training windows. Capacity and eligibility diagnostics are reported over 2010–2024, while model performance is evaluated over the available 2019–2024 walk-forward out-of-sample period.

The backtest uses the Section 6.3 cost schedule and the AUM-specific eligibility files. Portfolio returns are computed using true dollar PnL divided by total AUM:

$$R_t = \frac{\text{PnL}_t}{\text{AUM}}.$$

The reported net return includes commission, slippage, and borrow costs.



(a) 50M AUM (b) 250M AUM (c) 1B AUM

Figure 11: Cumulative strategy returns across portfolio AUM levels. Assumptions: maximum 200 long and 200 short names, zero-centered one-standard-deviation score threshold, and coursework-specified cost schedule.

6.3 Gross-to-Net Sharpe Decomposition

The final headline backtest follows the coursework-specified Section 6.3 cost schedule rather than the square-root diagnostic estimates in Section 3. Commission is charged at 1 bp of round-trip traded notional, auction slippage is charged at 3 bps of round-trip traded notional, and borrow is charged on the short book. Since the strategy trades close-to-open every day, the high daily turnover causes these fixed round-trip costs to consume most of the gross alpha.

Table 7: Sharpe Ratios Before and After Cost Components

AUM	Gross	After Comm.	After Slip.	Net
50M	1.12	0.49	-1.42	-1.53
250M	1.15	0.51	-1.41	-1.52
1B	1.30	0.67	-1.20	-1.30

Table 8: Sharpe Degradation by Cost Component

AUM	Comm. Pts	Slip. Pts	Borrow Pts	Total Pts
50M	0.64	1.91	0.11	2.65
250M	0.64	1.92	0.11	2.67
1B	0.63	1.88	0.10	2.60

The largest source of degradation is slippage, which reduces Sharpe by about 1.88–1.92 points. Commission reduces Sharpe by about 0.63–0.64 points, while borrow costs reduce Sharpe by about 0.10–0.11 points. Thus, the model has positive gross predictive value, but the daily close-to-open implementation is highly sensitive to trading costs.

The gross Sharpe remains positive across all three AUM levels, ranging from 1.12 at 50M to 1.30 at 1B. However, once commission, auction slippage, and borrow costs are applied, the net Sharpe becomes negative at all AUM levels. This shows that the signal has meaningful gross ranking value, but the short holding period and daily round-trip execution make the strategy difficult to monetize after realistic costs.

6.4 QuantStats

The 250M QuantStats tear-sheet benchmarks the strategy against $SP500\_TR$. Before costs, the volatility-matched comparison suggests that the strategy has positive gross alpha and can outperform the benchmark. However, after applying commission, auction slippage, and borrow costs, the performance deteriorates sharply.

Therefore, the main issue is not the absence of signal, but that the close-to-open alpha is too small relative to daily round-trip trading costs.

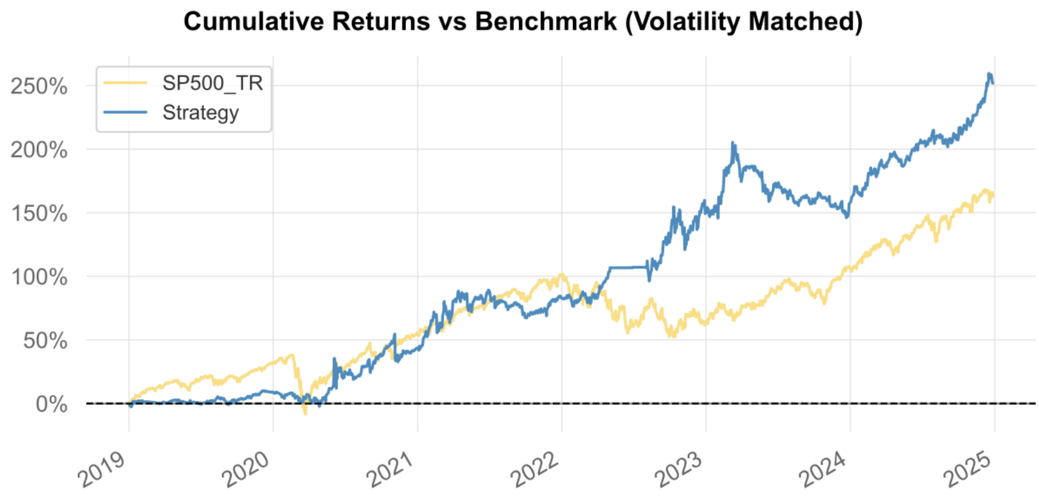


Figure 12: Cumulative returns of the gross 250M strategy versus $SP500\_TR$, with benchmark volatility matched to the strategy. Assumptions: maximum 200 long and 200 short names, before transaction costs.

6.5 Stress-Window Analysis

I evaluate the 250M strategy during three dislocation windows: 2020 Q1, the 2022 drawdown, and the March 2023 banking stress period. The results show that the strategy's gross signal is not uniformly defensive during market stress.

Table 9: 250M Strategy Performance in Stress Windows

Window	Days	Gross Ret.	Net Ret.	Gross Sharpe	Net Sharpe	Net MDD
2020 Q1 COVID shock	62	-0.23%	-1.90%	-0.46	-3.86	-2.37%
2022 drawdown	209	3.51%	-0.52%	1.88	-0.28	-1.64%
2023 banking stress	23	-0.48%	-1.28%	-1.48	-3.94	-1.81%

The strategy behaves best in the 2022 drawdown on a gross basis: it earns 3.51% with a gross Sharpe of 1.88, suggesting that the cross-sectional signal can still rank stocks well in a bear market. Still, after costs, the same window falls to -0.52%, so the edge is mostly absorbed by trading frictions.

The strategy performs poorly in sharper dislocations. In 2020 Q1, the gross return is already negative at -0.23%. The March 2023 banking stress window is similar: gross return is -0.48%, while net return is -1.28%. Overall, the model is more vulnerable in abrupt shock regimes, where overnight news, liquidity stress, and market-wide deleveraging dominate the stock-specific signals used by the random forest.

7. Conclusion

The main conclusion of this project is that the model does contain meaningful predictive information, but the close-to-open trading implementation is economically difficult to monetize after realistic costs. The final random forest signal achieves an out-of-sample IC mean of 0.02318 and an ICIR of 0.1561, which indicates that the factor combination has genuine cross-sectional ranking ability. In other words, the model is not simply fitting noise: it can rank stocks by their subsequent close-to-open returns with statistically meaningful accuracy.

However, the portfolio implementation is structurally expensive. Every position is opened at the closing auction and closed at the next opening auction, so each selected name pays a full round-trip cost every trading day. Unlike a lower-frequency daily strategy that can hold positions across multiple days and reduce turnover, this close-to-open strategy mechanically trades both legs every day. As a result, even a small per-leg cost becomes a large annual performance drag. The gross strategy has positive predictive value, but commission, auction slippage, and borrow costs consume most of the gross alpha.

The AUM constraint also makes the strategy harder to scale. At larger AUM levels, the strategy needs to trade larger dollar amounts in each stock while still respecting liquidity, ADV, eligibility, and borrow constraints. Under these restrictions, the investable basket becomes smaller and the realised gross exposure declines, especially at the 1B AUM level. This reduces the ability of the strategy to fully express the signal and further weakens net performance.

Therefore, my interpretation is that the factor model is useful, but the close-to-open portfolio format is not the most natural way to monetize daily and quarterly features. With more time and a broader factor-mining process, it may be possible to find stronger signals that survive the cost schedule and generate positive net performance. However, in general, overnight strategies are more naturally suited to event-driven signals, very short-horizon news information, or intraday/high-frequency order-book features. Daily factors, accounting variables, and quarterly fundamental signals are slower-moving by nature, so they are less suitable for a strategy that must fully enter and exit every position overnight.

Overall, the project shows a clear distinction between statistical predictability and tradable profitability. The model produces a valid cross-sectional signal, but the close-to-open execution structure imposes high daily turnover and heavy round-trip costs. The limiting factor is therefore not the absence of alpha, but the mismatch between the horizon of the features and the cost intensity of the trading rule.

8. Appendix

8.1 Additional Data Cleaning Details

Data Processing and Cross-Sectional Normalization

For each out-of-sample universe year, I construct the modelling dataset using the stocks in that year's point-in-time universe and include both the current year and the prior 12 months of observations. This gives enough history for lagged features, rolling transformations, validation, and model training while preserving the yearly universe definition.

Before model training, I apply a cross-sectional missingness filter. For each universe, a feature is retained only if at least 50% of stocks have non-missing values in each daily cross-section on average. This removes features that are too sparse to be reliable for daily cross-sectional prediction.

Next, I winsorize each feature cross-sectionally using a 5-MAD rule. For each date t and feature j, I compute the cross-sectional median $m_{t,j}$ and median absolute deviation $\mathrm{MAD}_{t,j}$, then clip the feature to $[m_{t,j} - 5\,\mathrm{MAD}_{t,j},\, m_{t,j} + 5\,\mathrm{MAD}_{t,j}]$. This reduces the effect of extreme outliers while preserving most of the ranking information.

After winsorization, missing values are filled using the same-date cross-sectional median. That is, if $X_{i,t,j}$ is missing after winsorization, I replace it with $m_{t,j}$. This keeps the imputation point-in-time and avoids using future information.

I then neutralize each feature cross-sectionally with respect to size and industry. For each date t and feature j, I run the regression

$$X_{i,t,j}^{\text{filled}} = \alpha_{t,j} + \beta_{t,j} \log(\mathrm{MktCap}_{i,t}) + \gamma_{t,j}^{\top} \mathrm{Industry}_{i,t} + \varepsilon_{i,t,j}.$$

The industry controls are GICS industry dummy variables, and the neutralized feature is the residual $X_{i,t,j}^{\text{neutralized}} = \varepsilon_{i,t,j}$. This removes common size and industry effects, so the model focuses more on stock-specific information rather than systematic market-cap or industry tilts.

Finally, I standardize each neutralized feature within the daily cross-section:

$$Z_{i,t,j} = \frac{X_{i,t,j}^{\text{neutralized}} - \mu_{t,j}}{\sigma_{t,j}}.$$

Here, $\mu_{t,j}$ and $\sigma_{t,j}$ are the same-date cross-sectional mean and standard deviation. The resulting features have approximately zero mean and unit variance within each date and are used as model inputs.

8.2 Additional Model Diagnostics

Table 10 reports additional diagnostics for the rolling random forest training procedure. Across 73 out-of-sample prediction windows, all 292 candidate model fits completed successfully. Each window evaluates four hyperparameter candidates and selects the best model using the validation long-short return.

Table 10: Additional random forest training diagnostics

Diagnostic	Mean	Min	Max
Raw training observations	104,998	104,985	105,000
Labeled training observations	42,103	42,081	42,105
Validation observations	21,000	20,986	21,000
Raw features before filtering	250	250	250
Features after permutation filter	150	61	235
Features removed by permutation	100	15	189
Initial feature clusters	85	81	90
Clusters after filtering / PCA features	60	37	76
Largest feature cluster size	48	48	48
Validation IC mean	0.0358	-0.0167	0.1125
Validation IC t-statistic	1.33	-0.97	4.15
Validation Sharpe	3.69	-6.63	16.54

The diagnostics show that the feature filtering step is active in every window. Starting from 250 raw features, the clustered permutation filter keeps about 150 features on average, and the final PCA representation contains about 60 model inputs. This reduces redundant and weak features while preserving the main correlated feature groups. The validation IC is positive on average, but its range also shows that some windows have weak or negative validation signal, which is consistent with the weaker out-of-sample regimes discussed in the main text.